Language, Computation and Cognition 096222

Project 1: Surprisal and RTs

Carmel Kenneth and Daniel Belousov

211785670 323929265

Abstract

This document contains our final project for the course. In the semi structured part, we chose to implement the GAM model bullet and the computation of RT-surprisal times bullet. In the Open section we chose to extend the findings in part 2 and train an RNN on dutch language, and analyze the results.

1 Introduction

We have chosen Project one because we connected more to the surprisal part of the course and found it to be interesting. Because of this we wanted to explore it more, and thus chose this path of the project.

2 Structured Section

We started by harmonizing the RNN-surprisal data with our reading time data and the N-gram surprisal's. This is done to get a computable table where the reading time, the N-gram surprisal's and the RNN surprisal's are aligned. (See Figure 1)

	code	token	wlen	surprisal	word	subject	text_id	text_pos	word_in_exp	time
0	17000	In	2	5.506053	In	s001	0	0	2285	399.90
1	17000	In	2	5.506053	In	s028	0	0	2503	290.32
2	17000	In	2	5.506053	In	s014	0	0	1394	501.59
3	17000	In	2	5.506053	In	s021	0	0	2525	210.93
4	17000	In	2	5.506053	In	s010	0	0	579	862.35
...	...	...	...	...	...	...	...	...	...	...
104133	35762	and	3	6.632089	and	s026	12	762	1501	317.57
104134	35762	and	3	6.632089	and	s008	12	762	4008	356.46
104135	35762	and	3	6.632089	and	s012	12	762	2159	420.24
104136	35762	and	3	6.632089	and	s021	12	762	1429	292.40
104137	35762	and	3	6.632089	and	s030	12	762	1488	303.85

104138 rows × 10 columns

Figure 1: Harmonized data frame

2.1 First Task

Using the Pearson test, we can get the correlations between the reading time and the RNN and N-gram surprisal estimations. We have received similar correlations between the read time and RNN surprisal and read time and N-gram surprisal of approximately 0.055 with the former having higher correlation of 0.0571, and the latter having a correlation of 0.0548. Hence RNN is slightly better than N-gram.

2.2 Second Task

We have plotted the relationship graph between the RNN surprisal (y axis) and N-gram Surprisal (x axis) (See figure 2). We see that there is a general "acceptance" between the models i.e., we receive a linear relationship like $y = x$. We can infer from the scales of graph that RNN predicts higher results than N-gram.

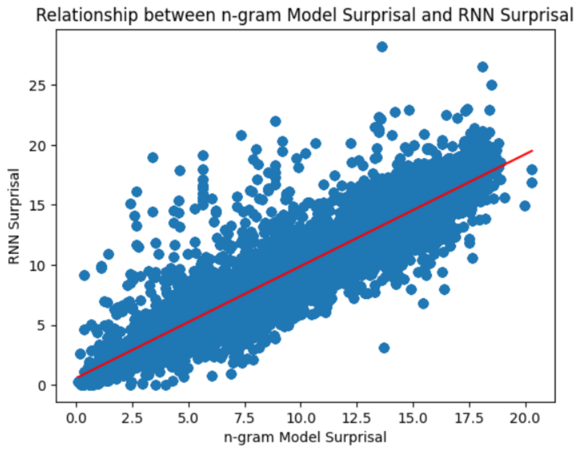

Figure 2: Rnn and N-gram relationship graph. With the red regression line.

2.3 Third Task

We wanted to find point where the models disagree with each other. We find (as we can also see if figure 2) that there are some "interest points" that RNN gives high surprisal when N-gram gives small surprisal, and we made a Pandas Data Frame containing some of these "interest point" – we got words like "john" and "the" that have different surprisals between the models. These words come from the following sentences:" *…was the John Harvey one…* " and *"…what to do The lifeboats were stuck…"*. (See figure 3). The data frame shows that its mainly RNN that has higher surprisal – at points – than N-gram. Of course, generally the models agree.

We think the models should disagree - for RNN can understand more complex language structures than N-gram thus should predict better.

	code	word	rnn surprisal	ngram surprise
0	17052	The	3.103844	13.6888

Figure 3: example of "interest point" in the data. In this case the point at (3.103844, 13.6888) – "the"

2.4 Fourth Task

To examine the spillover effect in both the Ngram and RNN models, we introduced a previous surprisal column to complement the existing surprisal column. By aligning the surprisal values with their respective probabilities, we constructed two arrays: x1, representing the probability values, and x2, representing the previous probability values. By plotting these arrays, we visualized the spillover effect and gained insights into how the probabilities of previous words impact the reading times of subsequent words.

In the analysis of spillover effects in the RNN model, it was observed that low to mid probabilities of the current word sometimes led to higher reading times for the next word. This finding suggests that when the current word is assigned a lower to moderate probability according to the RNN model, it can create a spillover effect where readers spend more time processing the following word.

Similarly, for the Ngram model, mid to high probabilities were also associated with higher reading times for the next word. This effect was particularly noticeable when the current word had a probability of 1.0. These results indicate that when the Ngram model assigns a high probability to the current word, it can create a spillover effect where readers spend more time processing the subsequent word.

However, it is worth to point out that the observed differences were visible but minor, these findings highlight the nuanced relationship between word probability and reading time in both the RNN and Ngram models. They suggest that even subtle variations in word predictability can have an impact on readers' processing dynamics. Further investigation is warranted to explore the underlying mechanisms and potential implications of these subtle spillover effects in language processing.

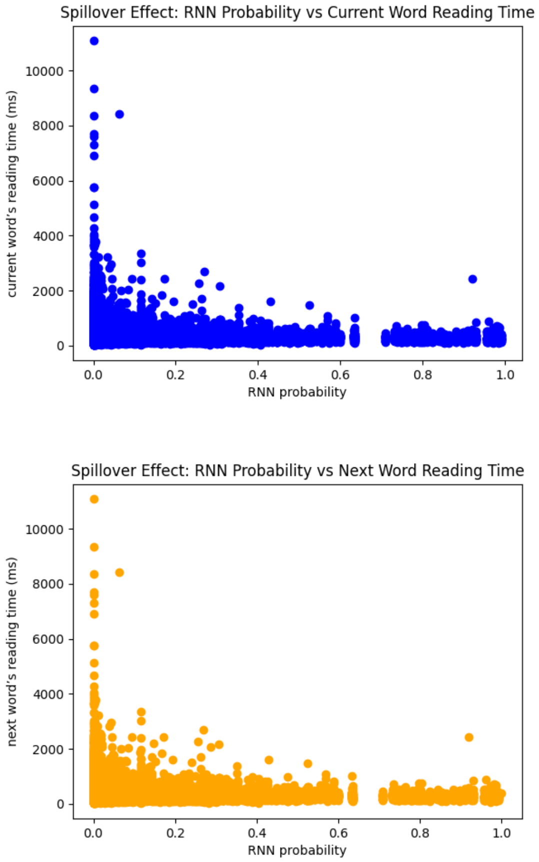

Figure 4: RNN spillover effect graph

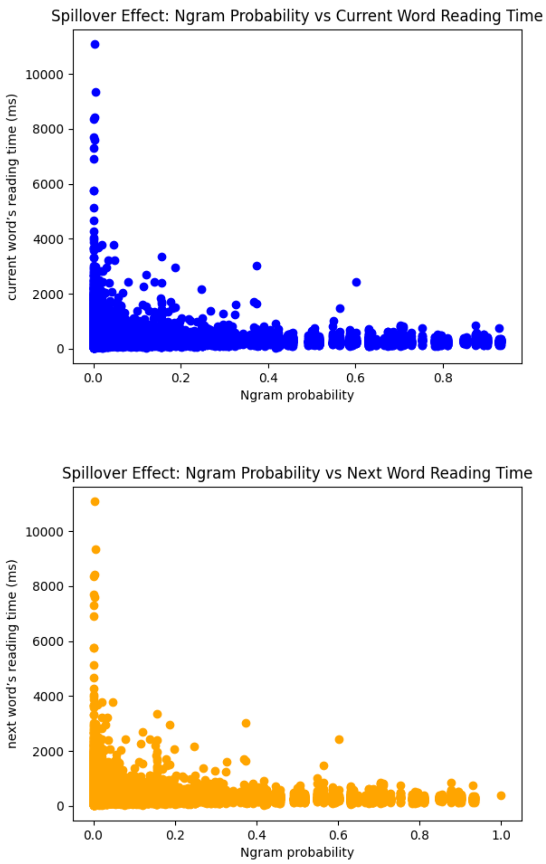

Figure 5: Ngram Spillover effect graphs

3 Semi Structured Task

As we stated in the abstract section, we chose to implement the first two bullets. i.e., we plotted the RT surprisal curve via the GAM model, and we chose the GECO eye tracking data base to compute surprisal's for it with the RNN from task one.

3.1 First Task

To build our GAM model, we first prepared a new dataset that incorporated the control variables for log-frequency and word length. Additionally, we introduced two additional columns: probability and previous probability, to specifically examine the spillover effect. To capture the relationship between the features and reading time, we constructed a linear GAM with 10 splines for each feature, except for the previous probability, where we used 5 splines. The decision to use a smaller number of splines for the previous probability feature was motivated by our intention to achieve a more linear approximation of its association with reading time. Subsequently, we fitted the GAM model to the reading time data and proceeded to visualize the results for the RNN model.

The RNN model includes four graphs that represent the marginal graphs of the GAM (Generalized Additive Model) concerning each feature: log-frequency of word, length of word, and surprisal of word. The model demonstrates that the GAM predicts longer reading times for words with higher log-frequency (we see some drops in values mid function), increased reading times for longer words, and elevated reading times for words with high surprisal (although we can see a drop in the reading times in the end of the curve for high surprisal values). These findings almost align with our intuition and the RT-Surprisal correlation that we have previously discussed (though the noticeable drop in the reading times for very high surprisal values warrants further investigation). Additionally, it becomes evident that word length has the most significant impact on reading time, which is an intuitive observation. However, for some reason no matter what we tried we got higher RT values for higher log-frequency which doesn't align with our initial expectations. This unexpected result could indicate a potential limitation or peculiarity in the dataset or the modeling approach. Further analyses are needed to understand the underlying factors contributing to this discrepancy.

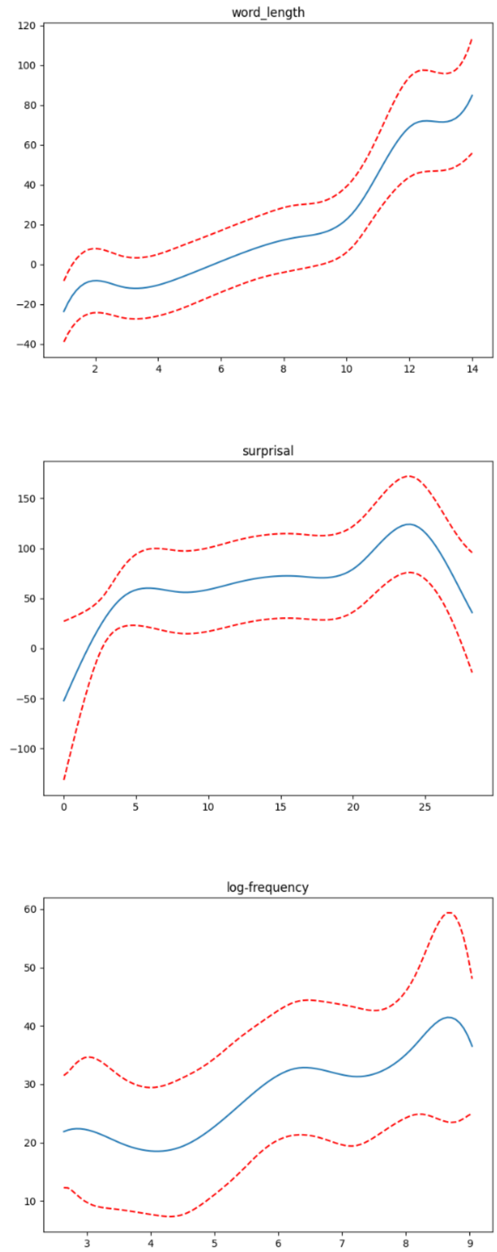

Figure 6: Curve graphs of the relationships of reading time with: word length, surprisal and log-frequency (from top to bottom)

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specifically examine the spillover effect. To capture the relationship between the features and reading time, we constructed a linear GAM with 10 splines for each feature, except for the previous probability, where we used 5 splines.

When examining the effects of both the current word and the spillover (preceding word) on reading time in the GAM model, we observed distinct patterns. Specifically, higher probabilities were consistently associated with lower reading times, indicating that more predictable words are processed more efficiently. However, when analyzing the spillover effect, we found a moderately concave relationship between the probability of preceding words and the reading time of subsequent words. This suggests that the influence of probability on the reading time of the next word is not as pronounced as the effect of the current word. These findings highlight the differential impacts of probability on reading time in the GAM model, with the current word effect showing a stronger association than the spillover effect.

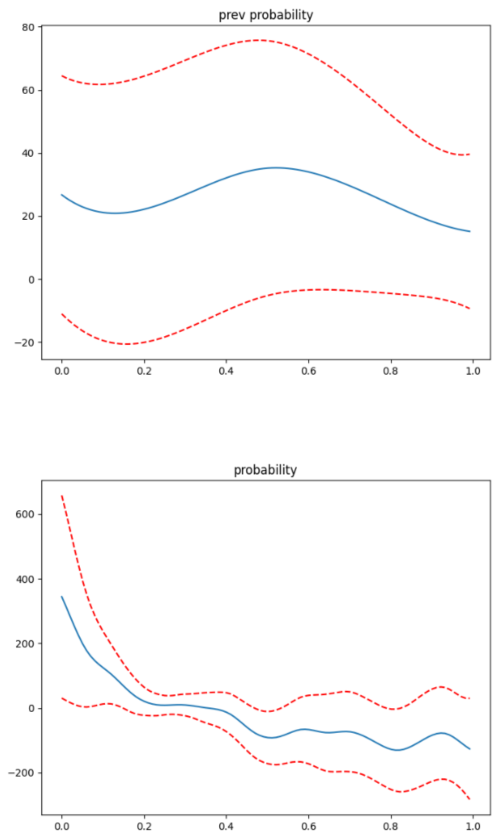

Figure 7: Curve graphs of the relationships of next word reading time with probability (upper graph) and the relationships of current word reading time with probability (lower graph).

3.2 Second Task

We chose to work with the GECO file. We extracted the data file from the given files in the task, and then we plugged it in to the RNN model to receive new surprisal's (See Fig 8). Now that we obtained the new surprisal's, we used scatter to plot to see the relationship between the new surprisal and the past surprisal (see in Fig 9). After this, we calculated the correlation between the new RNN surprisal, and the reading time, and we received a correlation of 0.5022. roughly the same as the original correlation.

Lastly, to examine the spillover effect in the RNN models, we introduced the previous surprisal column to complement the existing surprisal column. By aligning the surprisal values with their respective probabilities, we constructed two arrays: x1, representing the probability values, and x2, representing the previous probability values. By plotting these arrays, we visualized the spillover effect and gained insights into how the probabilities of previous words impact the reading times of subsequent words. (See Fig 10)

We suspect that in our RNN model that we trained and predicted on the GECO corpus we got different spillover effect then the structured. It appears that for some reason the effect is more noticeable in lower probabilities, where the lower the probability the lower the reading time.

	token	sentence_id	token_id	wlen	surp	entropy	entred
0	In	0	0	2	7.940023	9.049943	0.000000
1	<unk>	0	1	5	3.830694	7.861761	1.188182
2	County	0	2	6	18.172482	8.229176	0.000000
3	<unk>	0	3	5	3.748725	7.642810	0.586366
4	near	0	4	4	11.906780	7.645205	0.000000
...	...	...	...	...	...	...	...
7229	as	363	15	2	6.782777	7.615422	2.026868
7230	a	363	16	1	5.236154	7.101804	0.513618
7231	leader	363	17	6	17.555832	9.736368	0.000000
7232	and	363	18	3	4.887006	7.344543	2.391825
7233	<unk>	363	19	5	3.185401	8.693374	0.000000

Figure 8: new predicted surprisal's

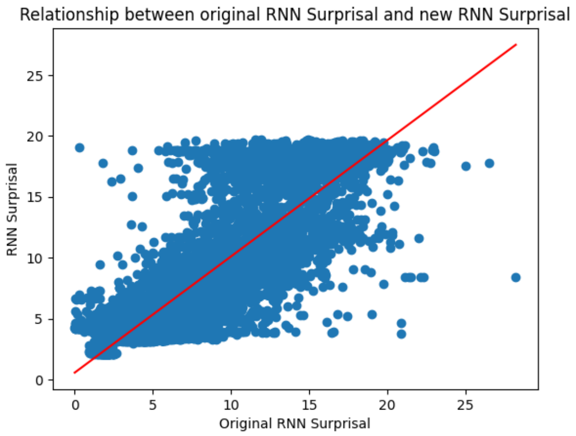

Figure 9: relationship between the surprisal's.

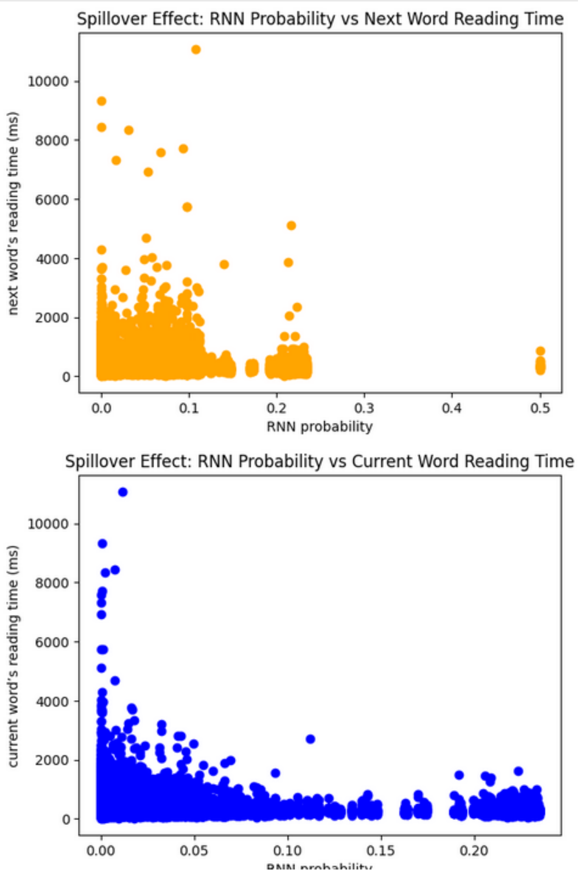

Figure 10: spillover effect with the new surprisals

4 Open Part

4.1 Introduction

In our open-ended task, we were inspired by the discussions on language models for other languages and decided to explore language modeling and analysis on Dutch. Our analysis focused on comparing the language model predictions from an RNN trained on sentences from the Netherlands parliament with the reading times of bilingual individuals, when the latter was received from the GECO dataset given to us in the class file. We aimed to understand how well the language model's predictions aligned with actual reading behavior, and explore the influence of surprisal, probability, and previous probability on reading times. By investigating these aspects, we aimed to gain insights into the interplay between language models and reading behavior.

4.2 Setup

For our analysis, we utilized a dataset consisting of 2 million sentences from the Netherlands parliament. Due to the large size of the dataset, we decided to reduce it to a more manageable size of 75,000 sentences. From this reduced dataset, we created a train set comprising 80% of the data and a validation set containing the remaining 20%.

To train our language model, we employed the RNN model on the new train data using a newly created vocabulary. The RNN model was trained for a total of 40 epochs, allowing it to learn from the data and improve its performance over time. We evaluated the model's performance on the validation set at each epoch to monitor its progress.

Using this trained RNN model, we computed the surprisal values for the reading time corpus. Each reading time was aligned with the corresponding word from the dataset, and the surprisal value was calculated based on the model's prediction for that word. This allowed us to quantify the level of surprise associated with each word in the reading time corpus.

To ensure data quality, we harmonized the reading times by removing corrupted or unknown rows. The dataset was further refined by aggregating the rows based on the mean reading time and eliminating duplicate entries. This resulted in a final dataset consisting of 2782 rows for our analysis.

To enrich our dataset, we incorporated additional features for each row. These features included the previous word probability, current word probability, and word frequency. By including these features, we aimed to explore their relationships with reading times and the language model's predictions.

Using this filtered and augmented dataset, we conducted our analysis.

4.3 Analysis

To begin, we examined the correlation between surprisal and reading times, aiming to determine the extent to which the language model's predictions influenced reading behavior. This analysis provided insights into the strength and direction of the association between these variables.

Moving forward, we focused on investigating the impact of word probability on reading times, considering both current word and spillover effects. We plotted the relationship between word probability and the reading time of the next word, exploring the potential "spillover" effect where the probability of the previous word influences the reading time of the subsequent word (See fig 11). Similarly, we analyzed the relationship between word probability and the reading time of the current word, shedding light on the influence of the current word's probability on reading behavior. By comparing these plots, we gained a nuanced understanding of the distinct effects of word probability on reading times.

To further explore the relationship between surprisal and reading times, we employed a General Additive Model (GAM). In the GAM model, we incorporated control variables such as log-frequency and word length to account for their potential impact on reading times, as well as examining both current word and spillover effects. Notably, we specifically chose the value of 5 for the number of splines (n_splines) in the GAM model. This choice was intended to simplify and linearize the relationships, allowing for a clearer interpretation of the patterns between surprisal and reading times.

4.4 Result

The results of the analysis provide insights into the relationship between various linguistic factors and reading time. Firstly, the computed surprisal values on the reading-time corpus indicated that words with lower probabilities (below 0.175) tended to have higher surprisal values, suggesting that these words were more unexpected or less predictable in the given context.

Examining the correlation between RNN surprisal and reading time revealed a weak positive relationship. This implies that as the surprisal value increased, indicating higher linguistic complexity or unpredictability, the reading time also tended to increase slightly. However, the correlation coefficient suggests that the relationship is relatively weak and other factors may have a more significant influence on reading time.

Analyzing the spillover effect, which investigates the impact of word probability on both the current word's reading time and the next word's reading time, yielded interesting findings. The correlation between probability and the current word's reading time was negative but weak, indicating a slight tendency for higher probabilities to be associated with lower reading times for the current word. However, the correlation between probability and the next word's reading time was very weak, suggesting that the influence of word probability on the subsequent word's reading time was negligible.

When visualizing the relationships, we observed an intriguing pattern in the relationship between word probability and reading times. Contrary to our initial expectation, lower probability words occasionally resulted in higher reading times for the next word. This finding suggests that when readers encounter words with lower probability, they may experience a momentary slowdown or increased processing effort, leading to longer reading times for the subsequent word. However, it is worth noting that this effect was not consistently observed across all lower probability words.

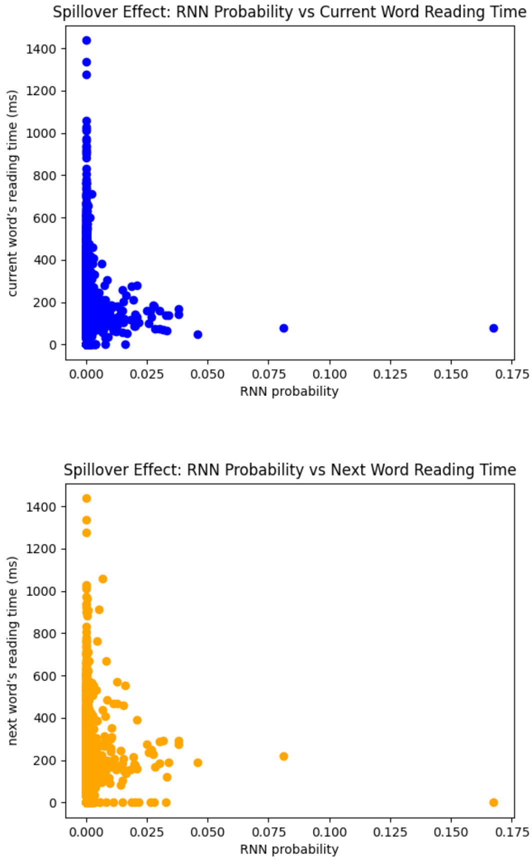

Figure 11: spillover effect graph

In the GAM model, log frequency exhibited a moderate convex relationship with reading time, indicating that less frequent words generally required more reading time. Word length showed a consistent positive linear relationship with reading time, suggesting that longer words took more time to process. Surprisal and probability demonstrated relatively stable lines, implying that their influence on reading time was minimal. The previous word's probability exhibited a slightly concave downward trend, indicating that higher probabilities of the previous word were associated with slightly shorter reading times for the current word.

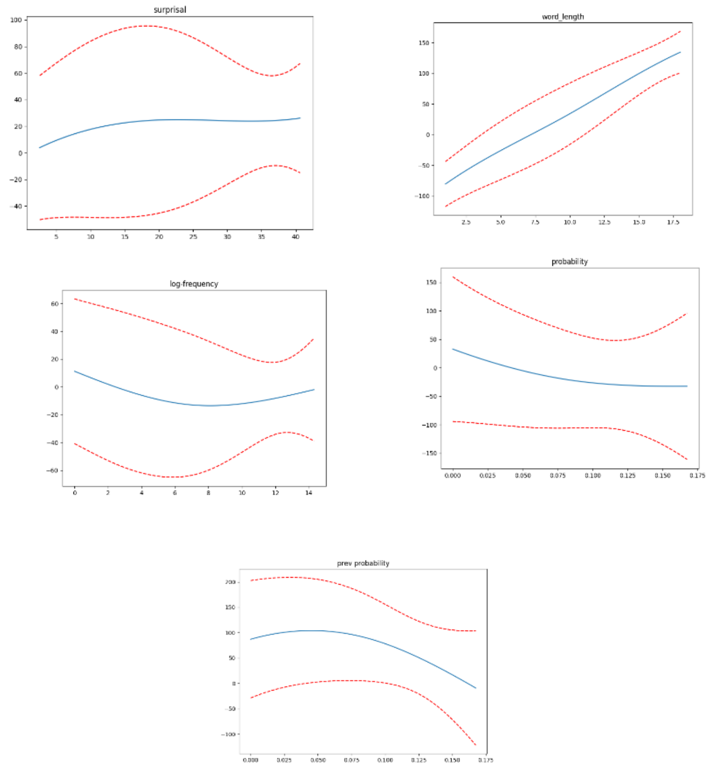

Figure 12: Relationship between Linguistic Factors and Reading Time in the GAM Model

4.5 Discussion

The results of the analysis provide insights into the relationship between various linguistic factors and reading time. Firstly, the computed surprisal values on the reading-time corpus indicated that words with lower probabilities (below 0.175) tended to have higher surprisal values, suggesting that these words were more unexpected or less predictable in the given context. This maybe due to the limited training set or poor model performance (maybe more epochs needed).

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In the GAM model, log frequency exhibited a moderate convex relationship with reading time, indicating that less frequent words generally required more reading time which is more logical compared to the RNN model from the structured task. Word length showed a consistent positive linear relationship with reading time, suggesting that longer words took more time to process. Surprisal and probability demonstrated relatively stable lines, implying that their influence on reading time was minimal, compared to the noticeable connection for these variables in the RNN model from the structured task. The previous word's probability exhibited a slightly concave downward trend, indicating that higher probabilities of the previous word were associated with slightly shorter reading times for the current word.

4.6 Summary

In summary, the analysis revealed interesting findings regarding the relationship between linguistic factors and reading time. Lower probability words were associated with higher surprisal values, indicating their unpredictability. The correlation between RNN surprisal and reading time was weak but positive, suggesting that greater linguistic complexity led to slightly longer reading times. The spillover effect analysis showed a weak relationship between word probability and reading time, with higher probabilities associated with lower reading times for the current word. Visualizations highlighted the intriguing effect of lower probability words on subsequent reading times. While the GAM model exhibited relationships between log frequency, word length, and reading time, surprisal and probability had minimal impact. Further optimization and exploration of other models and factors could provide additional insights into reading time dynamics.

References

1. Our google colab code
2. Dutch parliament protocols
3. GECO data base (including the English text for the semi structured second bullet, and the dutch text for the open section)